

Hadoop Installation and Verification

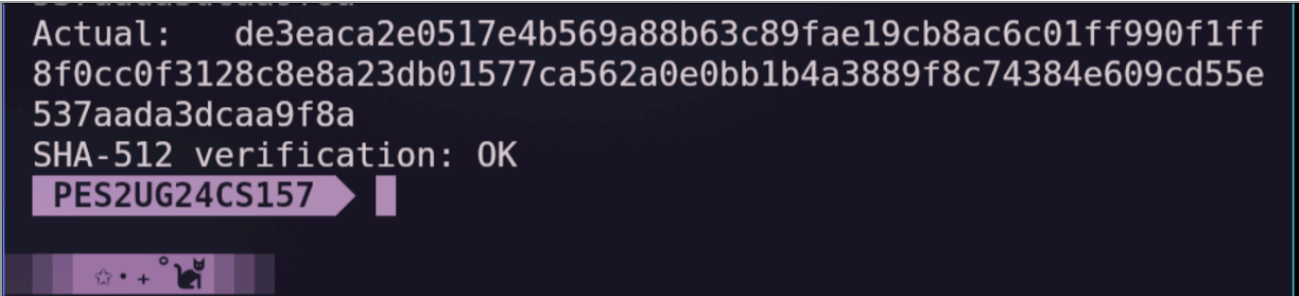
Name: Dharani S

SRN: PES2UG24CS157

Section: C

Hadoop Installation

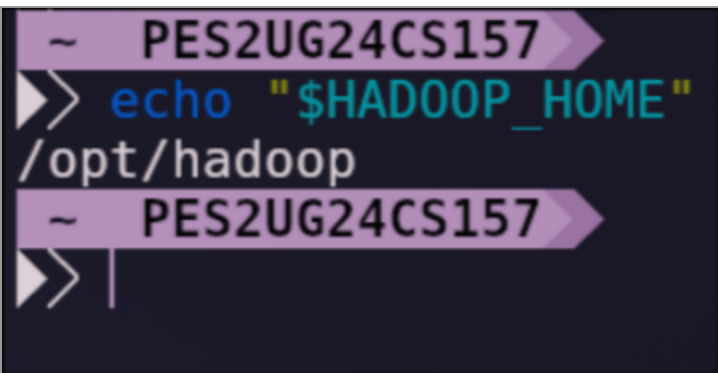
1. Download and verify Hadoop 3.3.6



```
Actual:   de3eaca2e0517e4b569a88b63c89fae19cb8ac6c01ff990f1ff
8f0cc0f3128c8e8a23db01577ca562a0e0bb1b4a3889f8c74384e609cd55e
537aada3dcaa9f8a
SHA-512 verification: OK
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```

A terminal window with a dark background. The first line shows a long hexadecimal string. The second line shows another long hexadecimal string. The third line shows a third long hexadecimal string. The fourth line shows 'SHA-512 verification: OK'. The fifth line shows the username 'PES2UG24CS157' followed by a prompt character. At the bottom of the terminal, there is a status bar with icons for a star, a plus sign, and a cat.

2. Environment Variables



```
~ PES2UG24CS157
> echo "$HADOOP_HOME"
/opt/hadoop
~ PES2UG24CS157
> |
```

A terminal window with a dark background. The first line shows the prompt '~ PES2UG24CS157'. The second line shows the command '> echo "\$HADOOP_HOME"'. The third line shows the output '/opt/hadoop'. The fourth line shows the prompt '~ PES2UG24CS157'. The fifth line shows the command '> |'.

```

~ PES2UG24CS157
>> sed -i "s|^# export JAVA_HOME=.*|export JAVA_HOME=$JAVA_HOME|" "$HADOOP_
CONF_DIR/hadoop-env.sh"

grep '^export JAVA_HOME=' "$HADOOP_CONF_DIR/hadoop-env.sh"
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
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```

3. Format HDFS (only once, reformatting wipes existing data)

```

node/current/fsimage_00000000000000000000.md5, /home/dharani/hadoop-data/nam
enode/current/edits_inprogress_00000000000000000120, /home/dharani/hadoop-da
ta/namenode/current/fsimage_00000000000000000119.md5, /home/dharani/hadoop-d
ata/namenode/current/edits_0000000000000000001-0000000000000000005]
2026-08-20 16:11:01,805 INFO common.Storage: Storage directory /home/dharan
i/hadoop-data/namenode has been successfully formatted.
2026-08-20 16:11:01,840 INFO namenode.FSImageFormatProtobuf: Saving image f
ile /home/dharani/hadoop-data/namenode/current/fsimage.ckpt_0000000000000000
0000 using no compression
2026-08-20 16:11:01,917 INFO namenode.FSImageFormatProtobuf: Image file /ho
me/dharani/hadoop-data/namenode/current/fsimage.ckpt_00000000000000000000 of
size 402 bytes saved in 0 seconds .
2026-08-20 16:11:01,935 INFO namenode.NNStorageRetentionManager: Going to r
etain 1 images with txid >= 0
2026-08-20 16:11:01,955 INFO namenode.FSNamesystem: Stopping services start
ed for active state
2026-08-20 16:11:01,956 INFO namenode.FSNamesystem: Stopping services start
ed for standby state
2026-08-20 16:11:01,959 INFO namenode.FSImage: FSImageSaver clean checkpoin
t: txid=0 when meet shutdown.
2026-08-20 16:11:01,959 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at server1/192.168.122.169
*****/
~ PES2UG24CS157
>> |

```

4. Start HDFS and YARN

```
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```

```
> start-dfs.sh  
start-yarn.sh
```

```
Starting namenodes on [localhost]
```

```
Starting datanodes
```

```
Starting secondary namenodes [server1]
```

```
Starting resourcemanager
```

```
Starting nodemanagers
```

```
~ PES2UG24CS157
```

```
> jps
```

```
10449 NameNode
```

```
10835 SecondaryNameNode
```

```
11043 ResourceManager
```

```
11383 NodeManager
```

```
11577 Jps
```

```
~ PES2UG24CS157
```

```
> |
```

Verification

1. HDFS health check

```
~ PES2UG24CS157
> hdfs dfsadmin -report
Configured Capacity: 61174157312 (56.97 GB)
Present Capacity: 46403751936 (43.22 GB)
DFS Remaining: 46403391488 (43.22 GB)
DFS Used: 360448 (352 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
r: 0
```

Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: server1
Decommission Status : Normal
Configured Capacity: 61174157312 (56.97 GB)
DFS Used: 360448 (352 KB)
Non DFS Used: 12139671552 (11.31 GB)
DFS Remaining: 46403391488 (43.22 GB)
DFS Used%: 0.00%
DFS Remaining%: 75.85%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Thu Aug 20 15:31:20 UTC 2026
Last Block Report: Thu Aug 20 15:28:26 UTC 2026
Num of Blocks: 3

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2. Basic HDFS read/write smoke test

```
PES2UG24CS157 ➤ hdfs dfs -mkdir -p /user/$USER  
hdfs dfs -mkdir /tmp-verify  
echo "hadoop verification line" | hdfs dfs -put - /tmp-verify/test.txt  
hdfs dfs -cat /tmp-verify/test.txt  
mkdir: `/tmp-verify': File exists  
put: `/tmp-verify/test.txt': File exists  
hadoop verification line  
PES2UG24CS157 ➤
```

3. YARN job, the pi example

```
2026-08-20 16:00:40,925 INFO impl.YarnClientImpl: Submitted application app
lication_1787239726166_0004
2026-08-20 16:00:40,959 INFO mapreduce.Job: The url to track the job: http:
//server1:8088/proxy/application_1787239726166_0004/
2026-08-20 16:00:40,960 INFO mapreduce.Job: Running job: job_1787239726166_
0004
2026-08-20 16:00:45,039 INFO mapreduce.Job: Job job_1787239726166_0004 runn
ing in uber mode : false
2026-08-20 16:00:45,041 INFO mapreduce.Job: map 0% reduce 0%
2026-08-20 16:00:49,112 INFO mapreduce.Job: map 100% reduce 0%
2026-08-20 16:00:54,180 INFO mapreduce.Job: map 100% reduce 100%
2026-08-20 16:00:55,210 INFO mapreduce.Job: Job job_1787239726166_0004 comp
leted successfully
2026-08-20 16:00:55,291 INFO mapreduce.Job: Counters: 54
```

```
2026-08-20 16:00:55,210 INFO mapreduce.Job: Job job_1787239726166_0004 comp
leted successfully
2026-08-20 16:00:55,291 INFO mapreduce.Job: Counters: 54
    File System Counters
        FILE: Number of bytes read=50
        FILE: Number of bytes written=830826
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=534
        HDFS: Number of bytes written=215
        HDFS: Number of read operations=13
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=3
        HDFS: Number of bytes read erasure-coded=0
    Job Counters
        Launched map tasks=2
```

Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=56
CPU time spent (ms)=1150
Physical memory (bytes) snapshot=887394304
Virtual memory (bytes) snapshot=8158760960
Total committed heap usage (bytes)=691011584
Peak Map Physical memory (bytes)=335372288
Peak Map Virtual memory (bytes)=2718932992
Peak Reduce Physical memory (bytes)=233607168
Peak Reduce Virtual memory (bytes)=2724536320

Shuffle Errors

BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters

Bytes Read=236

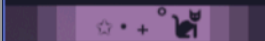
File Output Format Counters

Bytes Written=97

Job Finished in 15.495 seconds

Estimated value of Pi is 3.5000000000000000000000

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4. Exiting HDFS

```
PES2UG24CS157 ➤ stop-dfs.sh
stop-yarn.sh
jps
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [server1]
Stopping nodemanagers
Stopping resourcemanager
10126 Jps
PES2UG24CS157 ➤
```